Math 55 worksheet Monday, 02/02/2026

1. Prove that $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.

Solution: ($\Rightarrow$) Suppose that $A \subseteq C$ and $B \subseteq C$, and let x be an arbitrary element of $A \cup B$. By definition, we have $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$. We now do a proof by cases. For our first case, assume $x \in A$. Then since $A \subseteq C$ we have $x \in C$, which is what we wanted. For our second case, assume $x \in B$. Then since $B \subseteq C$ we get $x \in C$, as desired.

($\Leftarrow$) Suppose $A \cup B \subseteq C$. We have to show two things: $A \subseteq C$ and $B \subseteq C$.
For $A \subseteq C$, assume $x \in A$. Then, we have $x \in A \cup B$ by definition. But since $A \cup B \subseteq C$, we get $x \in C$. Thus, $A \subseteq C$. The proof for $B \subseteq C$ is similar.

2. Prove or disprove that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

Solution: We have to show two directions: first that $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$, and second that $A \times (B \cup C) \supseteq (A \times B) \cup (A \times C)$.

For $\subseteq$, let $(x, y) \in A \times (B \cup C)$. Then we know that $x \in A$ and $y \in B \cup C$. By definition, we know $y \in B \cup C \Leftrightarrow y \in B \vee y \in C$. Now we do cases. Suppose $y \in B$. Then, $(x, y) \in A \times B$, and hence $(x, y) \in (A \times B) \cup (A \times C)$. Now suppose $y \in C$. Then $(x, y) \in A \times C$, and so just like before we get $(x, y) \in (A \times B) \cup (A \times C)$.

For $\supseteq$, suppose $(x, y) \in (A \times B) \cup (A \times C)$. Then either $(x, y) \in A \times B$ or $(x, y) \in A \times C$. Suppose the first case is true. Then $x \in A$ and $y \in B$, and so $y \in B \cup C$. Hence, $(x, y) \in A \times (B \cup C)$. Now suppose the second case is true. Then $x \in A$ and $y \in C$, and so we get $y \in B \cup C$. Thus, we get $(x, y) \in A \times (B \cup C)$.

3. Prove or disprove that for all sets A and B, we have $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.

Solution: This is false; there exist sets A, B for which $\mathcal{P}(A \cup B) \neq \mathcal{P}(A) \cup \mathcal{P}(B)$. For example, consider $A = \{1\}$ and $B = \{2\}$. Then the set $\{1, 2\} \subseteq A \cup B$, and so $\{1, 2\} \in \mathcal{P}(A \cup B)$. But the set $\{1, 2\}$ is not a subset of either A or B, and hence is not an element of $\mathcal{P}(A) \cup \mathcal{P}(B)$.

4. Show that if a, b, x, y are elements, then $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$ if and only if $a = x$ and $b = y$. What methods of proof are most convenient to use for each direction?

Solution: ($\Leftarrow$) The backwards direction is easier, and since it's easier, we'll do that one first.
Suppose $a = x$ and $b = y$. Then we clearly get $\{a\} = \{x\}$ and $\{a, b\} = \{x, y\}$. Since an element z is in $\{\{a\}, \{a, b\}\}$ iff z is equal to either $\{a\}$ or $\{a, b\}$, and we have the above set equalities, we get $\{\{x\}, \{x, y\}\} \subseteq \{\{a\}, \{a, b\}\}$. The $\supseteq$ direction is similar. Note that we used a direct proof.

($\Rightarrow$) This direction is harder. We'll do a proof by contradiction. Suppose that $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$ but either $a \neq x$ or $b \neq y$. Suppose we're in the first case, where $a \neq x$. By hypothesis $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$, and so in particular since $\{a\} \in \{\{a\}, \{a, b\}\}$ we know that either $\{a\} = \{x\}$ or $\{a\} = \{x, y\}$. But since $x \neq a$, there's an element in both $\{x\}$ and in $\{x, y\}$ that's not

in $\{a\}$, and so $\{a\}$ cannot be equal to either of these sets.
Now consider the case where $b \neq y$. Similarly to last time, we know the set $\{a, b\}$ must either be equal to $\{x\}$ or to $\{x, y\}$. If $\{a, b\} = \{x\}$, then we get $a = b = x$. But then, the set $\{x, y\}$ is not equal to either $\{a\}$ or $\{a, b\}$ because y is not in either of them. So we must be in the case where $\{a, b\} = \{x, y\}$. By hypothesis, we have $y \neq b$, so we must have $y = a$ and $b = x$. But then we must be in our earlier case where $a \neq x$, which is a contradiction.